

OPEN PROJECT 2026

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

Using Large-Scale Charging Session Data

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Data Landscape & Preprocessing

ACN-Data

Adaptive Charging Network

- 30,000+ EV sessions
- Caltech & JPL sites
- Timestamps, energy delivered
- Session duration & user IDs
- JSON → CSV converted

UrbanEV (ST-EVCDP)

Shenzhen Urban Grid

- 24,798 charging piles
- 5-minute interval data
- Large-scale urban charging
- Temporal & spatial patterns
- Peak-hour analysis

Missing values handled using interpolation and forward-fill based assumptions.

ENGINEERED FEATURES

2,134,080 records | 38 features | Master Dataset v2

Utilization Rate

Revenue per Session

Queue Proxy

Congestion Score

Peak/Off-Peak Period

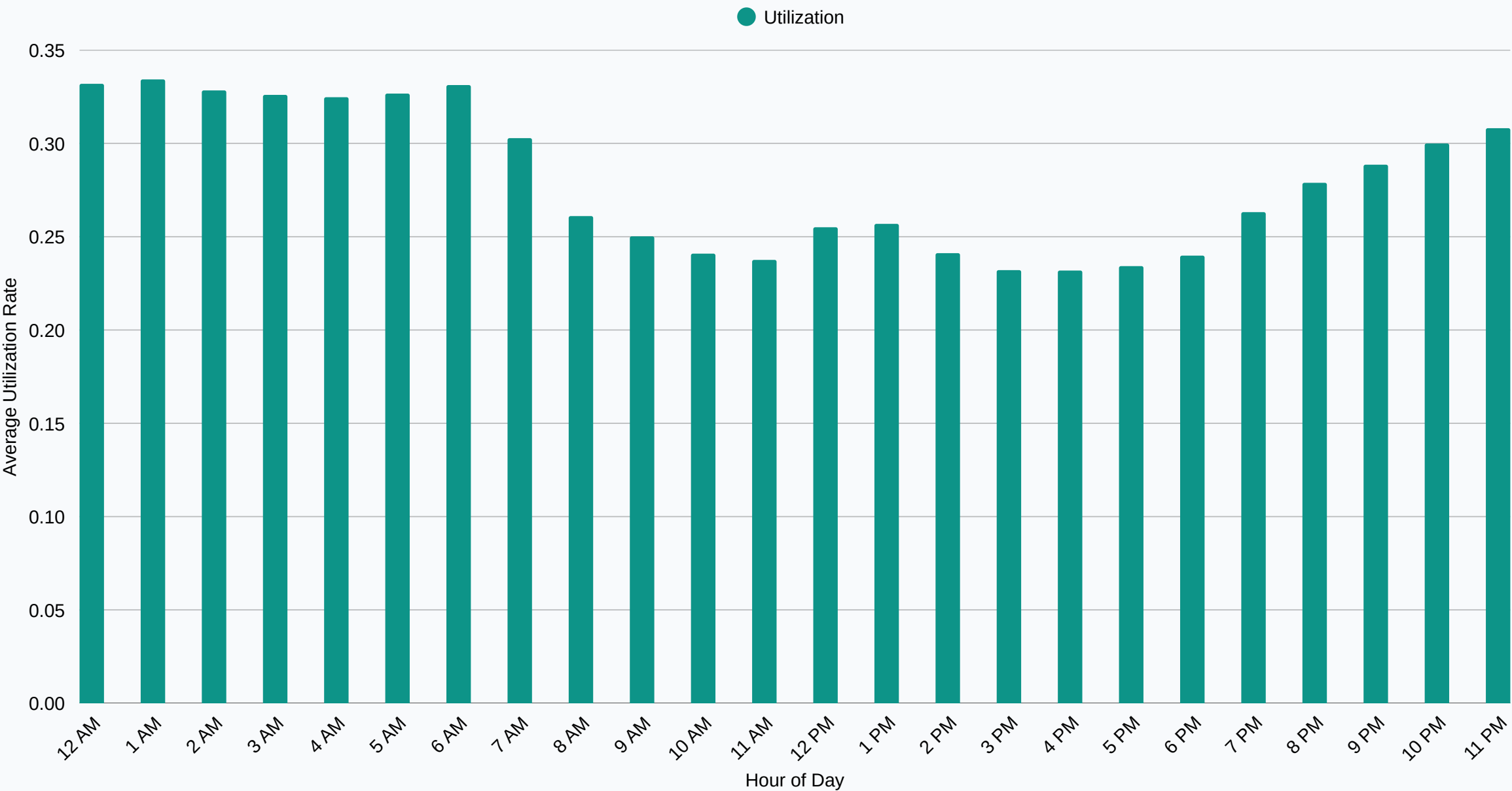
User Type Proxy

Profit per kWh

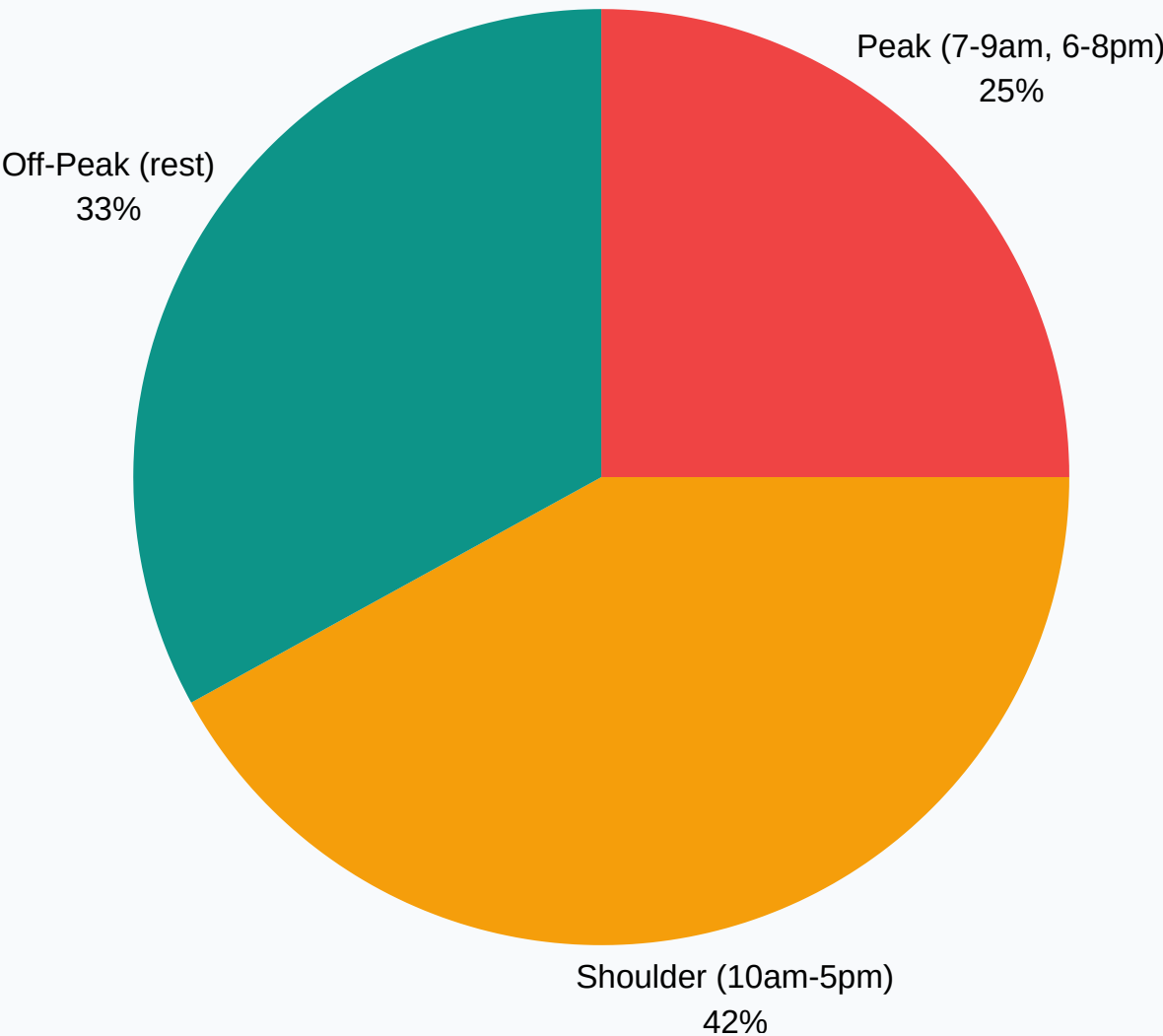
Charger Mix Ratio

Key EDA Findings & Demand Behavior

Average EV Charger Utilization by Hour of Day



Charging Session Distribution Across Peak, Shoulder and Off-Peak Periods



65%

Sessions in peak/shoulder hours

28%

Avg baseline utilization rate

34.5%

Fleet users

9am & 7pm

Twin peak congestion hours

Fleet users identified using high utilization + long duration proxy.

Peak demand windows indicate ideal candidates for surge pricing.

Demand Prediction Agent

0.9933

R² Score

Explains 99.3% variance

0.0068

MAE

Mean Absolute Error

0.0144

RMSE

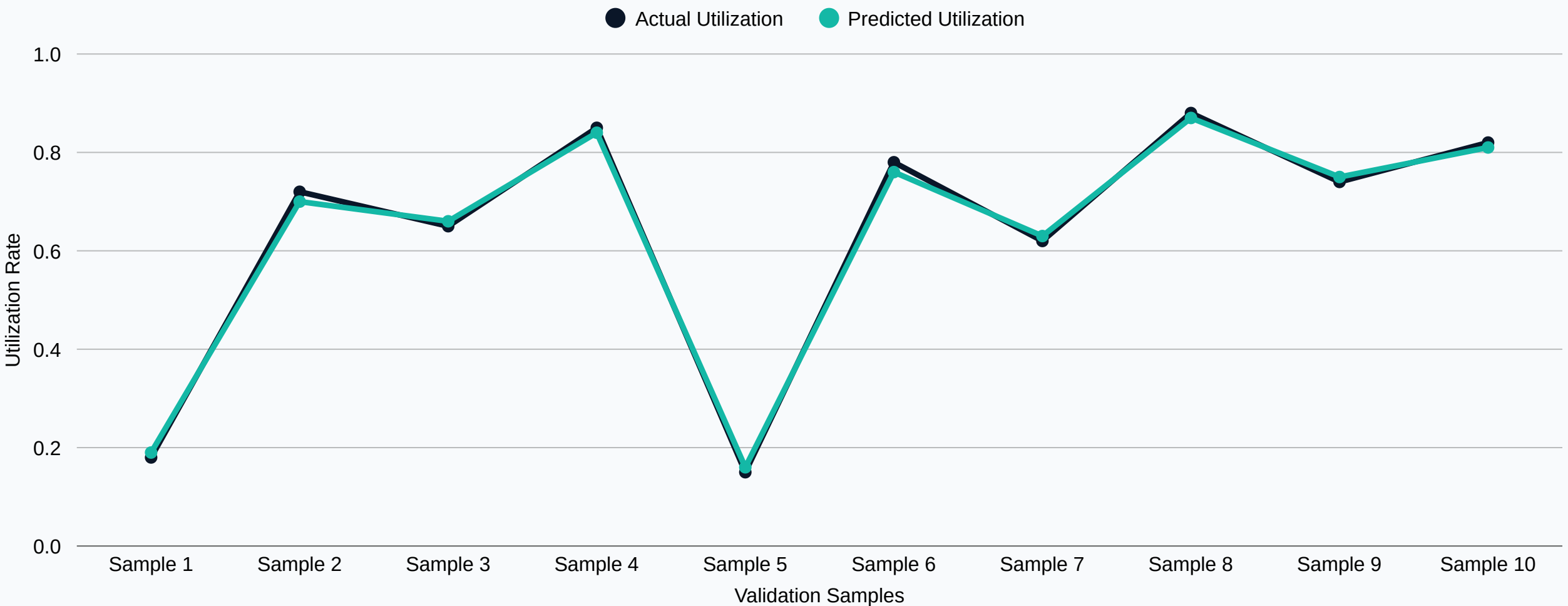
Root Mean Squared Error

Average Predicted Utilization
0.280028

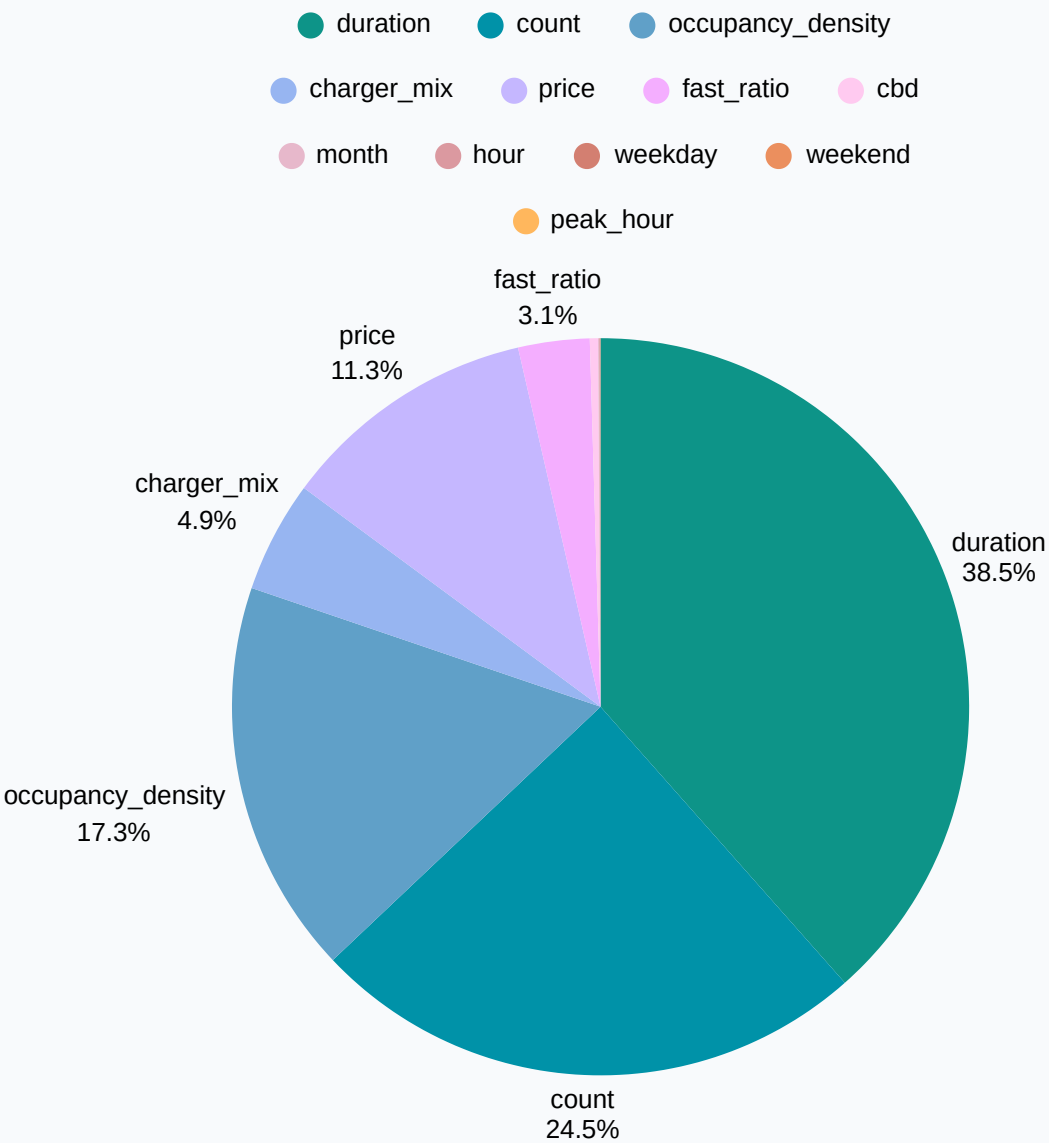
Average Expected Charging Load
21.892364

Average Congestion Probability
0.349074

Actual vs Predicted Charger Utilization(Random Forest)



Feature Importance Ranking for Demand Prediction Model



The predicted utilization values closely track actual utilization values, demonstrating strong model accuracy ($R^2 = 0.9933$) and low prediction error.

Dynamic Tariff Pricing Agent

SURGE PRICING

₹18/kWh

Utilization > 80%
(kWh ≥ Q75: 13.2 kWh)

STANDARD

₹15/kWh

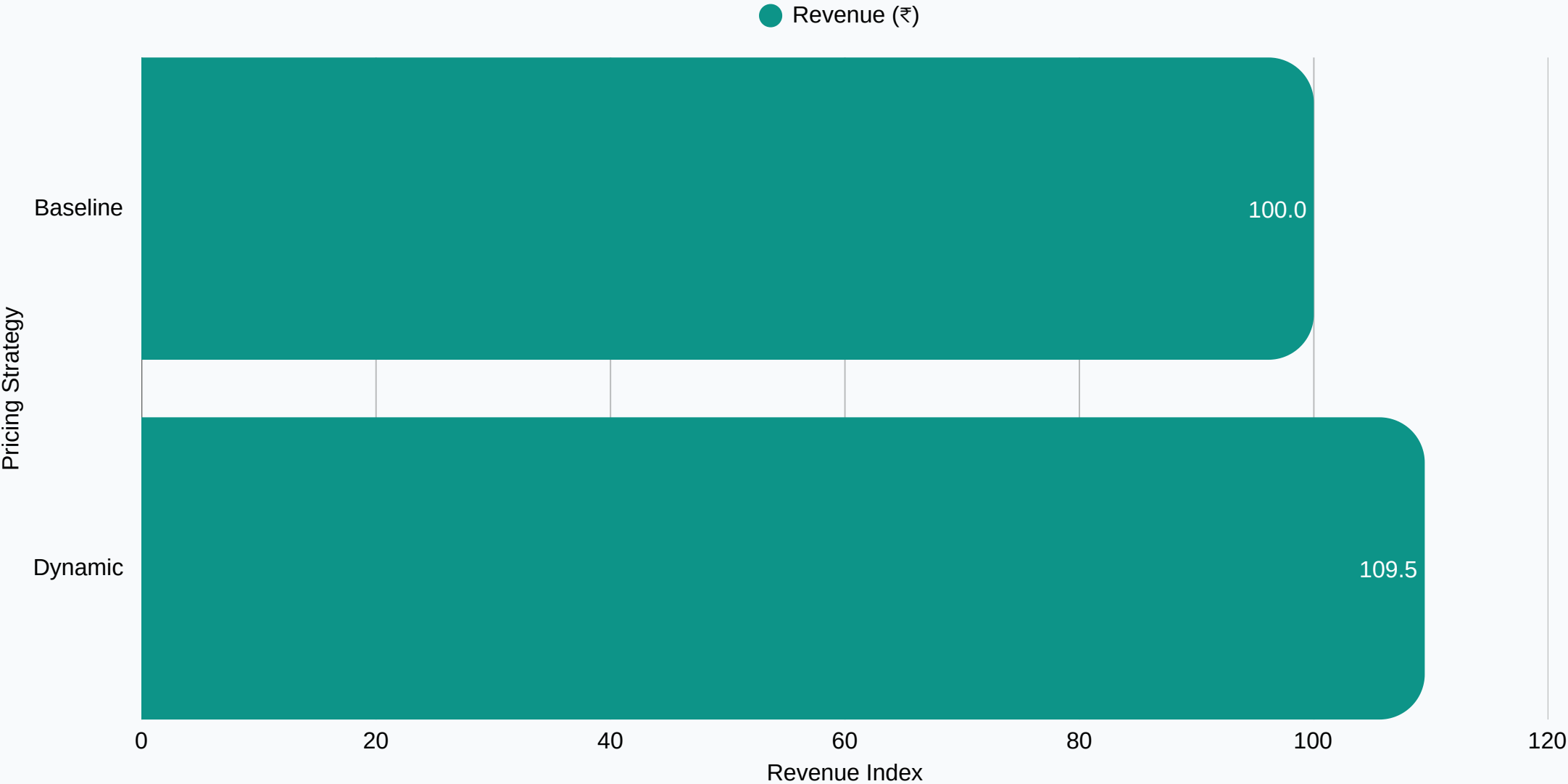
Normal Demand
(Q25 < kWh < Q75)

DISCOUNT

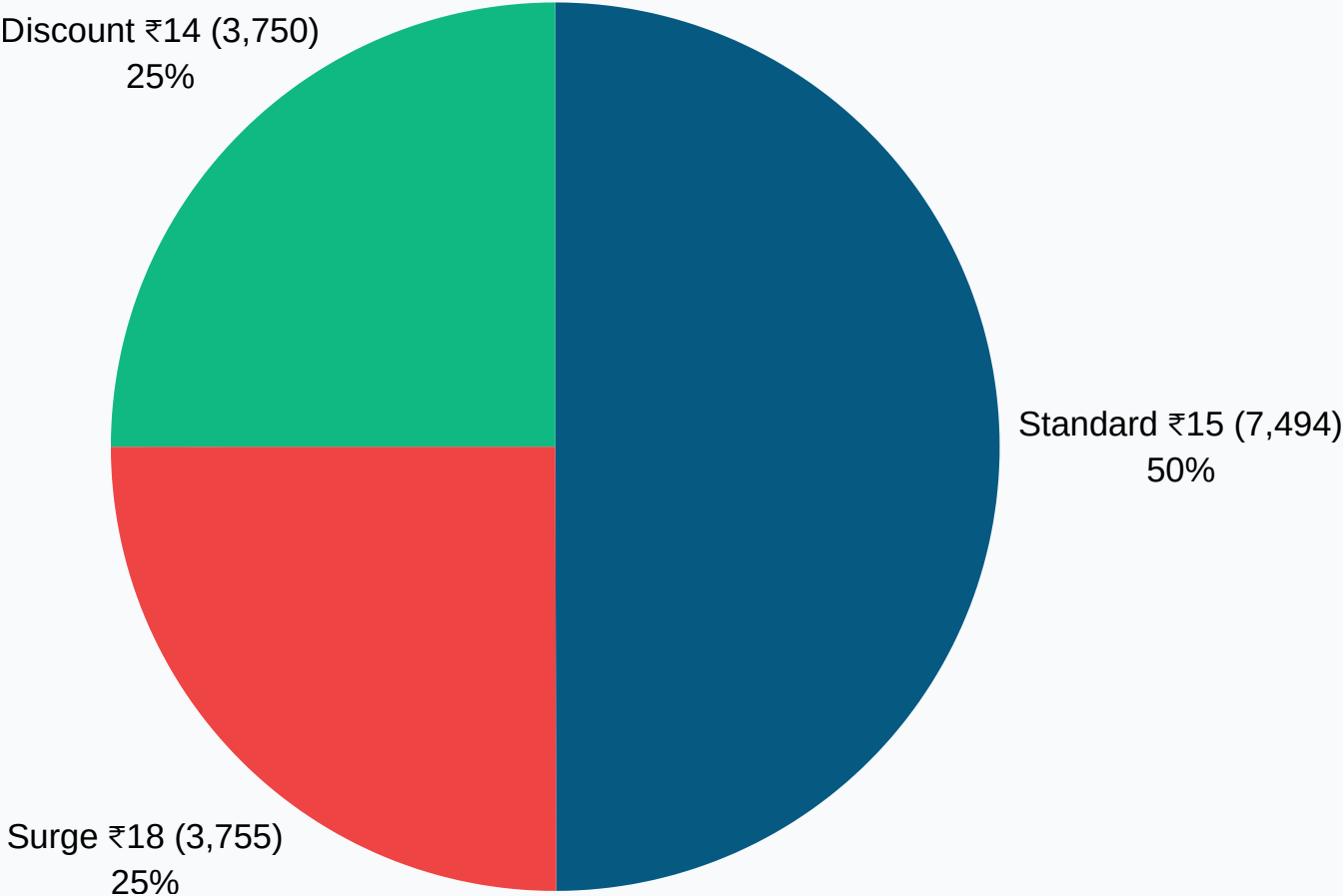
₹14/kWh

Utilization < 30%
(kWh ≤ Q25: 4.0 kWh)

Revenue Index: Fixed Tariff vs Dynamic Pricing



Dynamic Tariff Recommendations
Generated by Pricing Agent



Monitoring & Learning Agent

Agent	Metric	Value	Interpretation
Demand Prediction	R ² Score	0.9933	Near-perfect demand prediction
Demand Prediction	MAE / RMSE	0.0068 / 0.0144	Very low prediction error
Tariff Pricing	Revenue Gain (%)	+9.46%	vs ₹15/kWh fixed baseline
Tariff Pricing	Off-Peak Uplift (%)	+20.0%	Discount pricing drives off-peak sessions
Monitoring	Wait Time Reduction (%)	5.0%	Peak queue reduction
ACN / Monitoring	Pricing Efficiency Score	₹16.38/kWh	Revenue per kWh delivered
ACN / Monitoring	Customer Response Rate (%)	0.23%	Net demand shift from elasticity

🔄 Feedback-driven monitoring loop evaluates pricing outcomes and provides signals for future tariff optimization.

Business, Operational & Policy Implications



Revenue Uplift

9.46% revenue gain over flat-rate pricing with dynamic tariff. Surge pricing on heavy users ($\geq$ Q75 kWh) adds significant margin without demand destruction.



Grid Balancing

Discount pricing (₹14) at off-peak hours drives 20% session uplift, reducing peak-hour load and grid strain, supporting demand-response compliance.



User Experience

5% reduction in peak wait times via congestion-aware pricing. Fleet users (34.5% of base) can be offered preferential off-peak rates for long-term loyalty.



Autonomous Intelligence

Self-improving feedback loop, monitoring agent evaluates every pricing decision in real time, enabling continuous learning without human intervention.



Policy Alignment

Dynamic tariffs incentivize off-peak EV usage aligned with India's green energy policies, reducing fossil dependency during peak grid hours.



Scalability

Framework tested on 24,798 piles (UrbanEV) + 15,000 ACN sessions. Architecture scales to any EV network with session-level data.

Results Summary & Final KPIs

99.33%

Demand Prediction R²

+9.46%

Revenue Gain (Dynamic vs Fixed)

+20%

Off-Peak Session Uplift

₹16.38

Pricing Efficiency (per kWh)

Utilization Metric	Value
Baseline Utilization Rate (Before Pricing)	28.02%
Post-Pricing Utilization Rate	28.01%

Utilization remained almost stable while revenue increased by 9.46%.

Key Assumptions

- Baseline tariff: ₹15/kWh fixed
- Discount → +20% demand uplift
- Surge → -2% demand reduction
- Queue proxy used for wait time
- Energy cost = ₹7/kWh

Raw Data
(ACN + UrbanEV)



Feature
Engineering



Demand
Prediction



Tariff
Pricing



Monitoring
& Learning



Revenue &
Efficiency

THANK YOU!